DAY 81

Part 1: Create Web API with EF Core

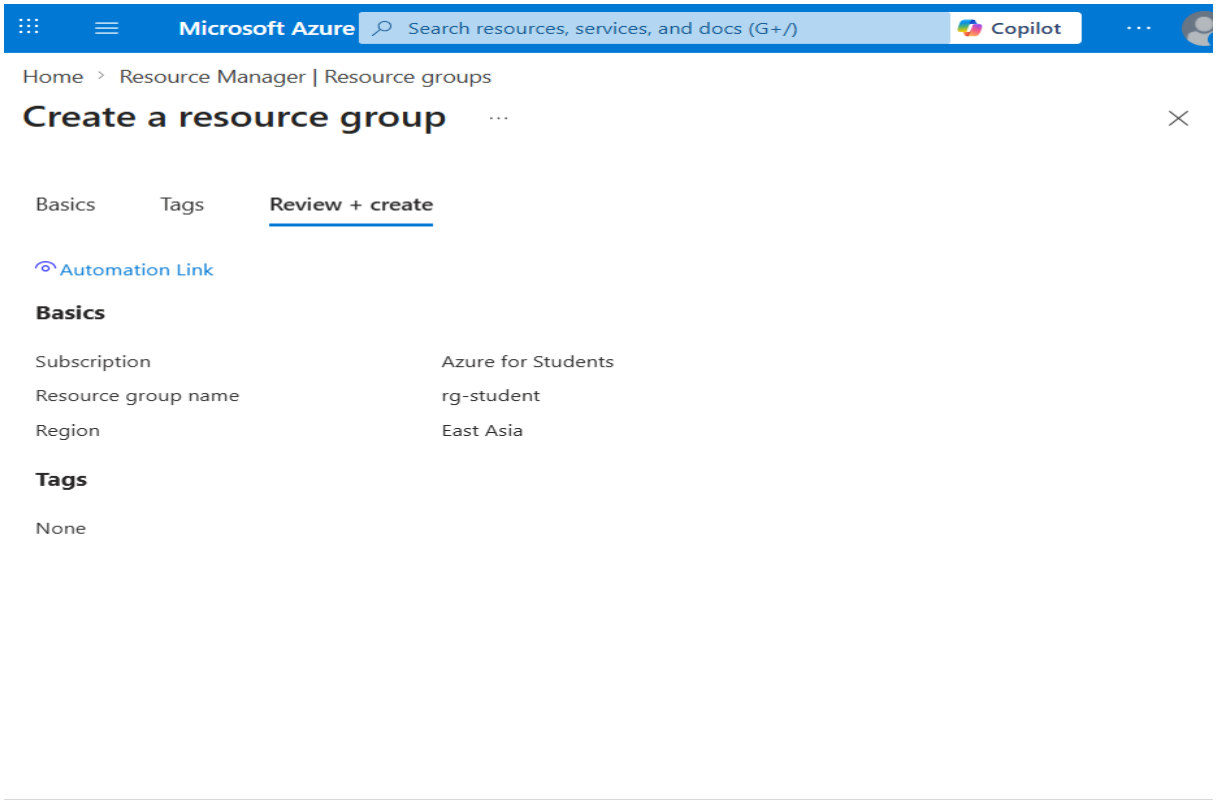
```
namespace WebAPIEFcore.Models
{

    public class Student
    {

        public int Id { get; set; }

        public string Name { get; set; }

        public int Age { get; set; }

        public string Course { get; set; }

    }
}
```

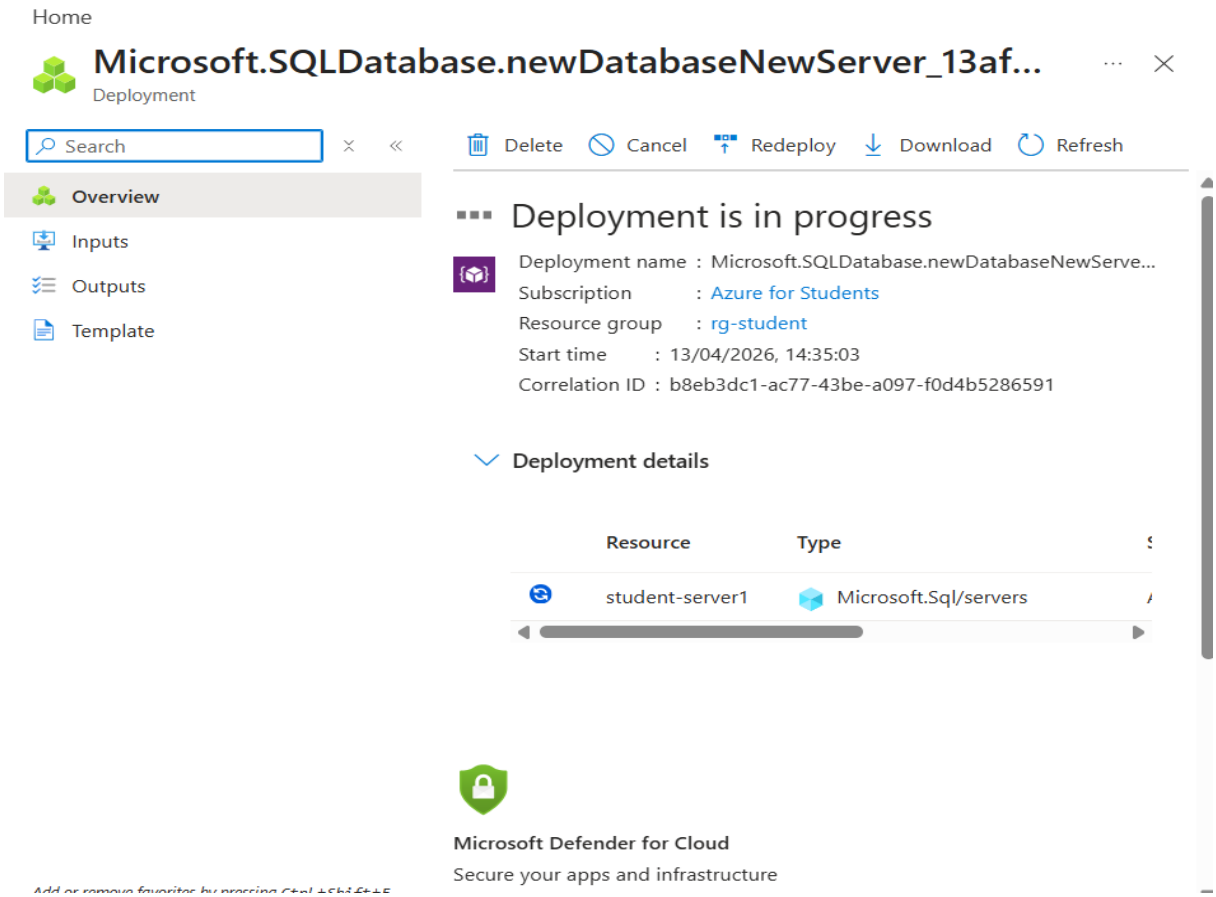
```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Threading.Tasks;
using Microsoft.AspNetCore.Http;
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using WebAPIEFcore.Data;
using WebAPIEFcore.Models;

namespace WebAPIEFcore.Controllers
{
    [Route("api/[controller]")]
    [ApiController]

    public class StudentsController : ControllerBase
    {
        private readonly WebAPIEFcoreContext _context;
```

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Threading.Tasks;
using Microsoft.EntityFrameworkCore;
using WebAPIEFcore.Models;

namespace WebAPIEFcore.Data
{

    public class WebAPIEFcoreContext : DbContext
    {

        public WebAPIEFcoreContext (DbContextOptions<WebAPIEFcoreContext> options)
            : base(options)
        {
        }

```

```
using Microsoft.EntityFrameworkCore.Migrations;

#nullable disable

namespace WebAPIEFcore.Migrations
{
    /// <inheritdoc />

    public partial class init : Migration
    {
        /// <inheritdoc />

        protected override void Up(MigrationBuilder migrationBuilder)
        {
            migrationBuilder.CreateTable(
                name: "Student",
                columns: table => new
                {
                    Id = table.Column<int>(type: "int", nullable: false)
```

Part 2: Create Azure SQL Database





Home

Microsoft.SQLDatabase.newDatabaseNewServer_13afaf4989324...

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_13afaf4989324fc...
Subscription : Azure for Students
Resource group : rg-student
Start time : 13/04/2026, 14:35:09
Correlation ID : b8eb3dc1-ac77-43be-a097-f0d4b5286591

Deployment details

Next steps

Go to resource

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Add or remove favorites by pressing Ctrl+Shift+F

Home > Microsoft.SQLDatabase.newDatabaseNewServer_13afaf4989324fcebb595 | Overview

student-db (student-server1/student-db)

SQL database

Suggest missing index | +2

Search

- Overview
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- Diagnose and solve problems
- Query editor (preview)
- Mirror database in Fabric (preview)
- Resource visualizer
- Settings
- Data management
- Integrations
- Power Platform
- Security
- Intelligent performance
- Monitoring
- Automation

Copy | Restore | Export | Set server firewall | Delete | ...

Essentials JSON View

Resource group (move)	: rg-student
Status	: Online
Location	: East Asia
Subscription (move)	: Azure for Students
Subscription ID	: 1713f309-616d-4a48-857a-27184b5f46ab
Server name	: student-server1.database.windows.net
Connection strings	: Show database connection strings
Pricing tier	: Free - General Purpose - Serverless: Gen5, 2 vCores
Overage billing	: Disabled
Free monthly vCore a...	: 100,000 vCore seconds remaining
Earliest restore point	: No restore point available
Tags (edit)	: Add tags

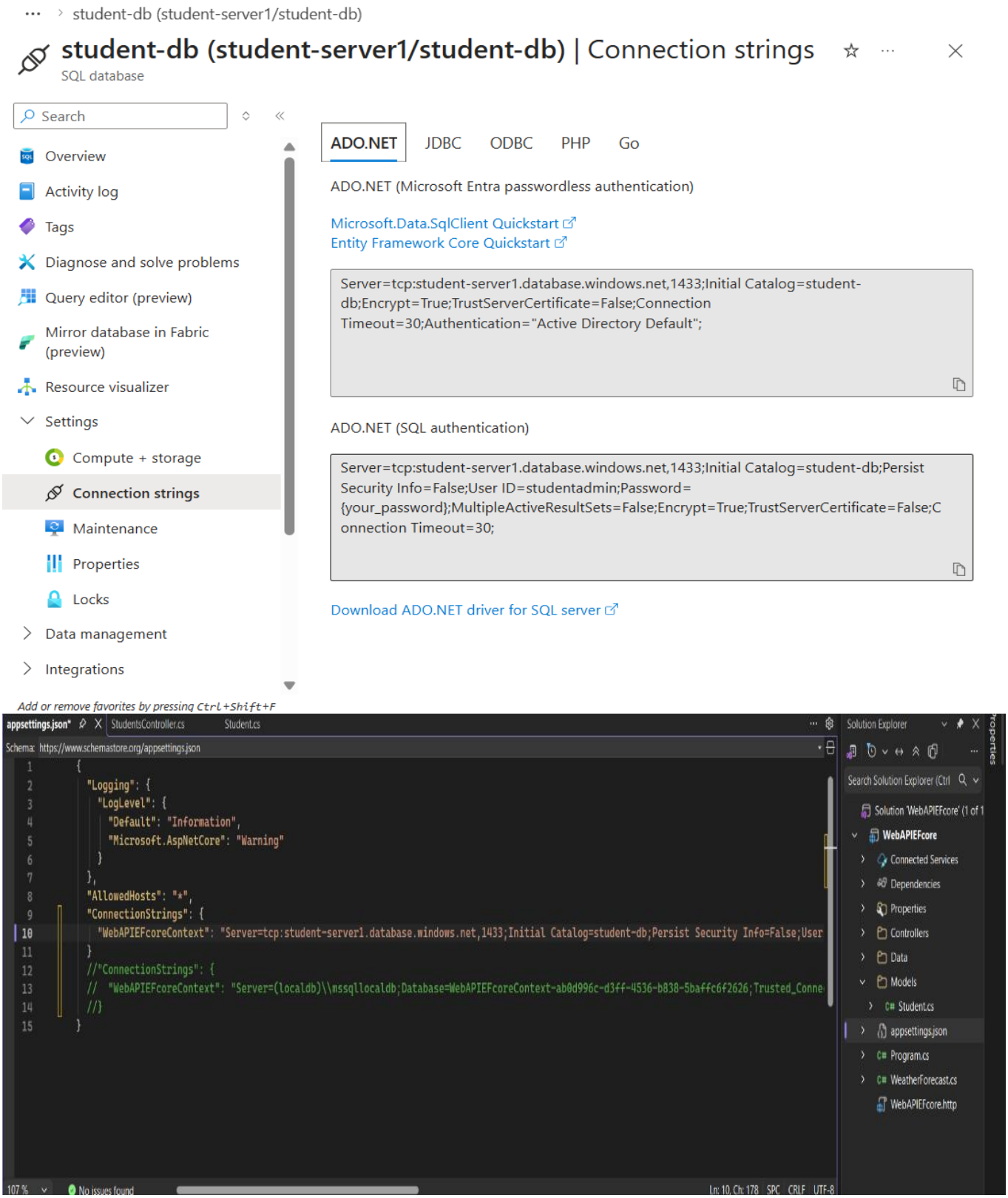
Getting started | Monitoring | Properties | Features | Notifications (0) | ...

Start working with your database

Connect to your database and start working with data with a few simple steps. Learn

Add or remove favorites by pressing Ctrl+Shift+F

Part 3: Secure Connection String



Microsoft Azure | Search resources, services, and docs (G+/) | Copilot

student-db (student-server1/student-db) > student-server1

student-server1 | Networking

SQL server

Search

- Overview
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- Data management
- Security
  - Networking
  - Microsoft Defender for Cloud
  - Transparent data encryption
  - Identity
  - Auditing

Allow virtual networks to connect to your resource using service endpoints. Learn more

+ Add a virtual network rule

Rule	Virtual netw...	Subnet	Address r...	Endpoint sta...	Resource
No virtual network rules found.					

Firewall rules

Allow certain public internet IP addresses to access your resource. Learn more

+ Add your client IPv4 address (45.248.159.10) + Add a firewall rule

Rule name	Start IPv4 address	End IPv4 address
ClientIPAddress_2026-4-13_1...	45.248.159.10	45.248.159.10

Exceptions

☑ Allow Azure services and resources to access this server

Save | Discard

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Home > student-db (student-server1/student-db)

student-db (student-server1/student-db) | Query editor (preview)

SQL database

New query | Templates | Open in | Feedback | Sign out | Classic experience

Search

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- Mirror database in Fabric (preview)
- Resource visualizer
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  - Connection strings
  - Maintenance
  - Properties
  - Locks
- Data management

Explorer

Search

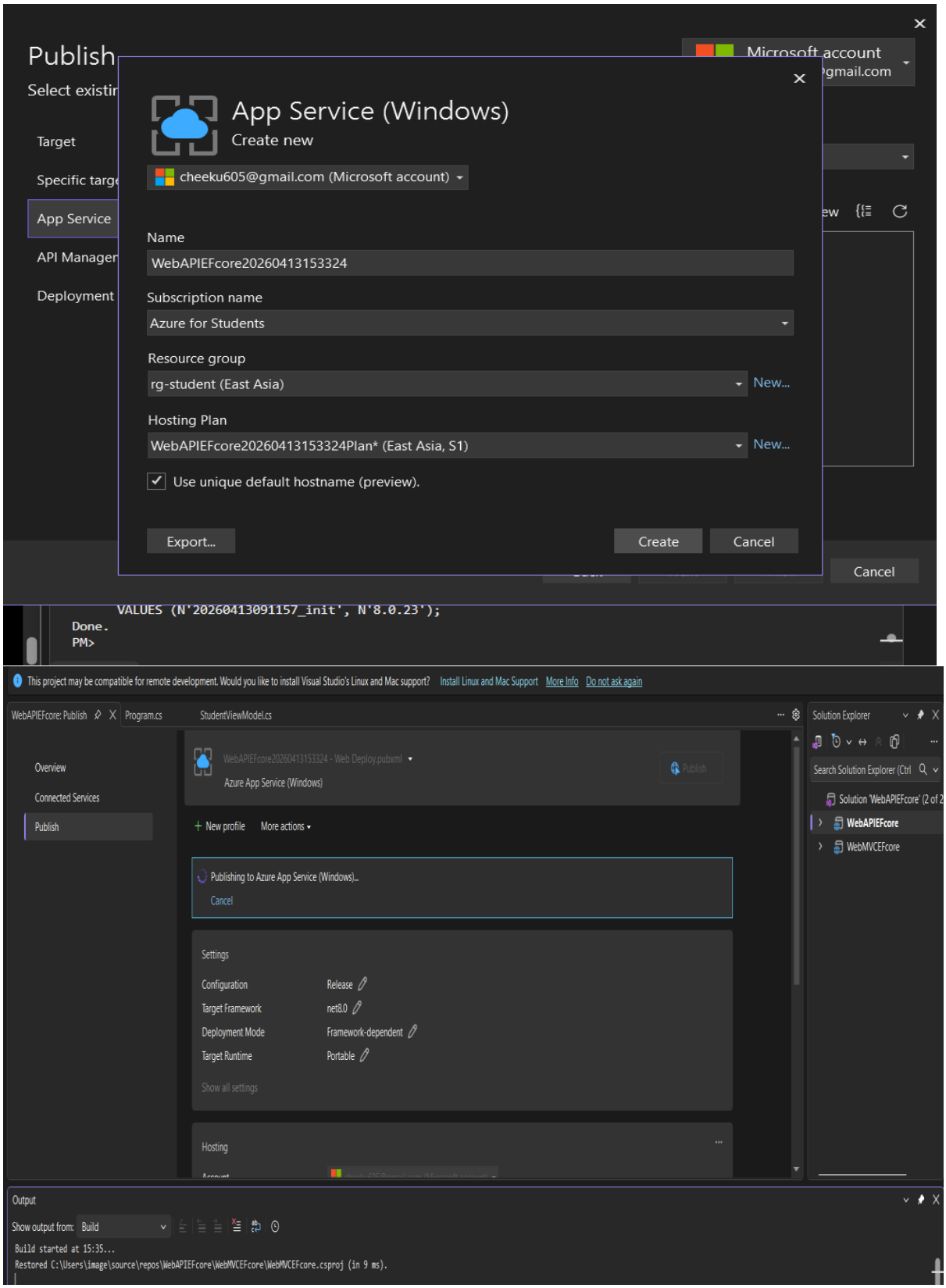
- student-db
  - dbo
    - Tables
      - _EFMigrations
      - Student
    - Views
    - Stored Procedures
    - Functions
  - Queries

Start exploring your data

Write and run SQL queries, create database objects with a template, or open a table or view to preview your data.

New query

Part 4: Deploy Web API to Azure



Part 5: Create MVC Application

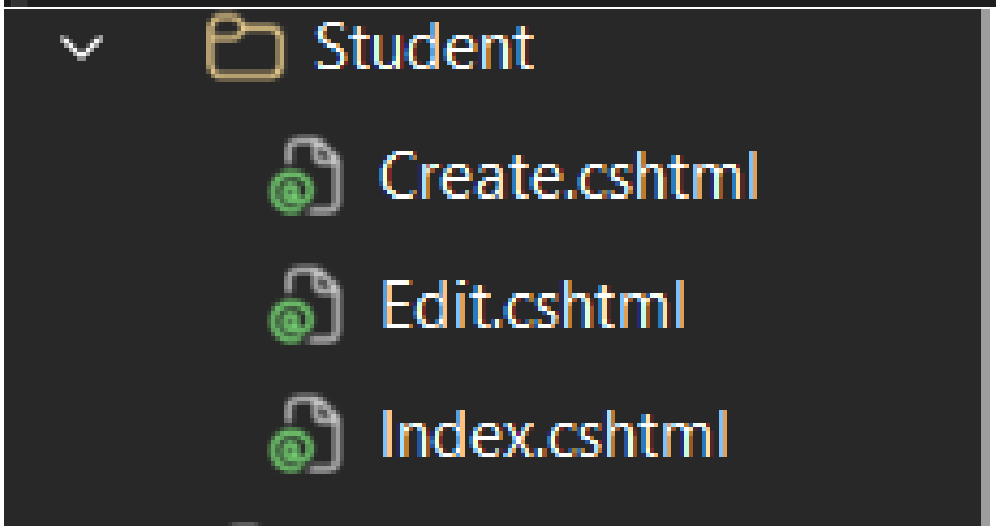
WebMVCEFcore | StudentController | Edit(int id)

```
using Microsoft.AspNetCore.Mvc;
using System.Text;
using System.Text.Json;
using WebMVCEFcore.Models;

1 reference
public class StudentController : Controller
{
    private readonly IHttpClientFactory _httpClientFactory;

    0 references
    public StudentController(IHttpClientFactory httpClientFactory)
    {
        _httpClientFactory = httpClientFactory;
    }

    // GET: Students
    0 references
    public async Task<IActionResult> Index()
    {
        var client = _httpClientFactory.CreateClient("API");

        var response = await client.GetAsync("api/students");

        if (!response.IsSuccessStatusCode)
            return View(new List<StudentViewModel>());

        var json = await response.Content.ReadAsStringAsync();
        var students = JsonSerializer.Deserialize<List<StudentViewModel>>(json,
            new JsonSerializerOptions { PropertyNameCaseInsensitive = true });

        return View(students);
```

WebMVCEFcore: Publish | Index.cshtml | Program.cs | Create.cshtml | StudentController.cs | WebAPIEFcore: Publish | StudentVie

WebMVCEFcore | WebMVCEFcore.Models.StudentViewModel | Name

```
namespace WebMVCEFcore.Models
{
    11 references
    public class StudentViewModel
    {
        4 references
        public int Id { get; set; }
        3 references
        public string Name { get; set; }
        3 references
        public int Age { get; set; }
        3 references
        public string Course { get; set; }
    }
}
```



student-db (student-server1/student-db) | Query editor (preview)

SQL database

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Replicas

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New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

Search

student-db
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Tables
__EFMigratio
Student
Colum...
Indexes
Views
Stored Proced...
Functions
Queries

SQL query 1

Run | Save as view

Your queries will be lost when your session

```
1  SELECT TOP (1000) * FROM [dbo].[Student]
```

Messages | Results

	Id	Name	Age	
1	1	Sneha Pahuja	21	DSA

Succeeded (0 sec 220 ms) Columns: 4 Rows: 1

webapiefcore20260413153324-gxgdggh4buamfdg7.eastasia-01.azurewebsites.net/api/Students

Google Meet Calls | Active Courses | CU-DotNet-Jan26-B4 | Email - SNEHA PAH... | NeoPAT | Strivers A2Z DSA Co... | UI/UX Design For B... | snehapahuja/CU-D...

Pretty print

[{"id":1,"name":"Sneha Pahuja","age":21,"course":"DSA"}]

localhost:7087

Google Meet Calls | Active Courses | CU-DotNet-Jan26-B4 | Email - SNEHA PAH... | NeoPAT

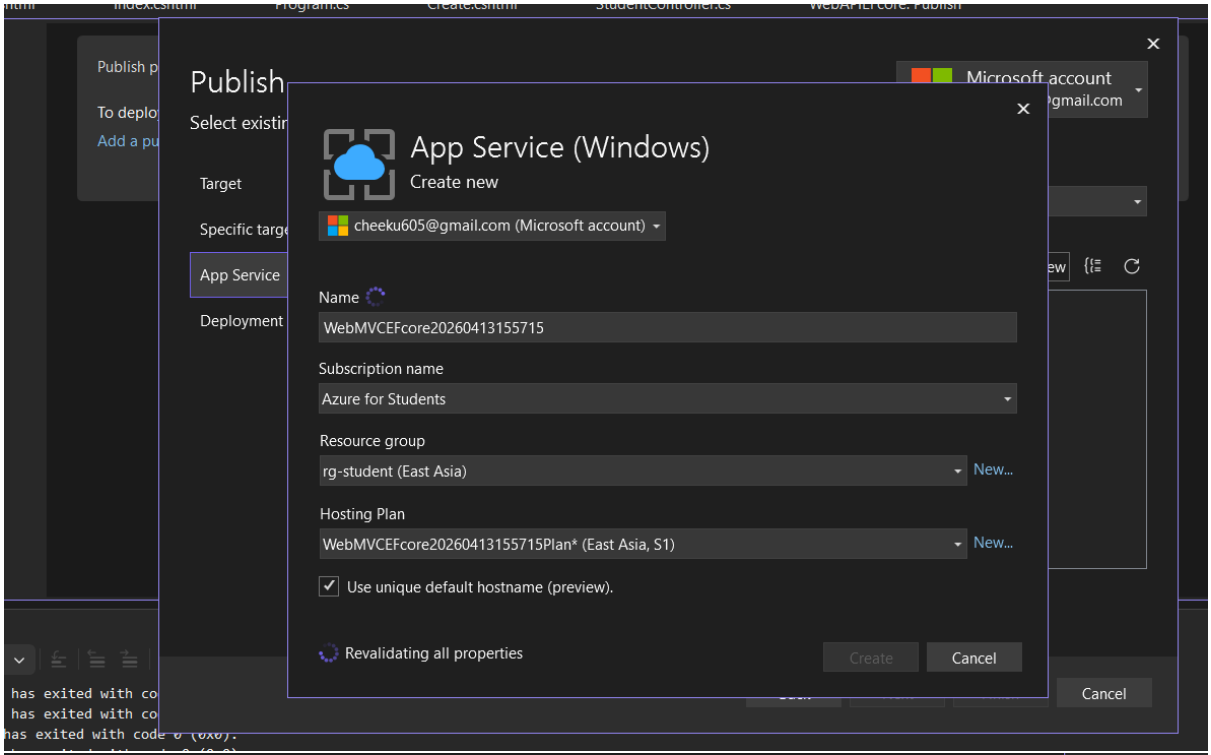
WebMVCEFcore Home Privacy

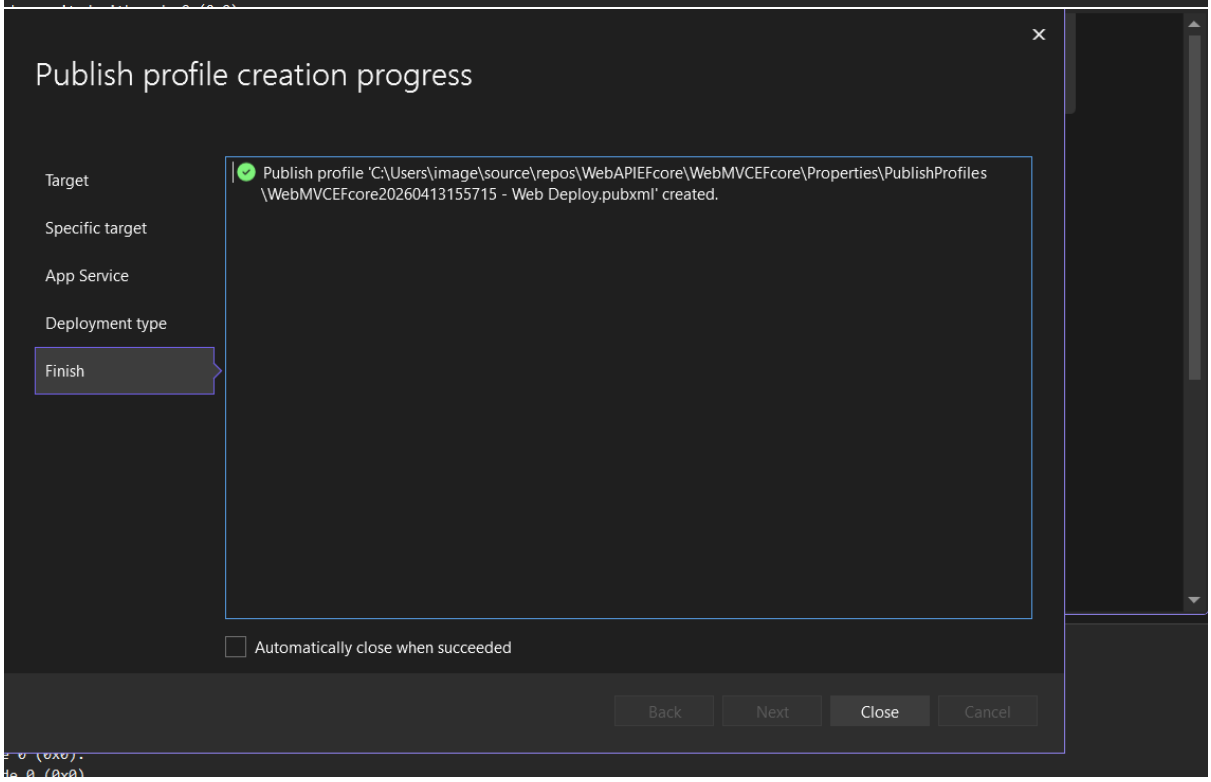
Students

Add New

Name	Age	Course	Actions
Sneha Pahuja	21	DSA	Edit | Delete

Part 6: Deploy MVC App





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WebMVCEFcore Home Privacy

Students

Add New

Name	Age	Course	Actions
Sneha Pahuja	21	DSA	Edit | Delete

WebMVCEFcore Home Privacy

Students

Add New

Name	Age	Course	Actions
Sneha Pahuja	21	DSA	Edit | Delete
angel	21	WEBDEV	Edit | Delete

Home > Azure SQL | SQL databases > student-db (student-server1/student-db)

Azure SQL | SQL databases

student-db (student-server1/student-db) | Query editor (preview)

SELECT TOP (1000) * FROM [dbo].[student]

Id	Name	Age	
1	Sneha Pahuja	21	DSA
2	angel	21	WEB

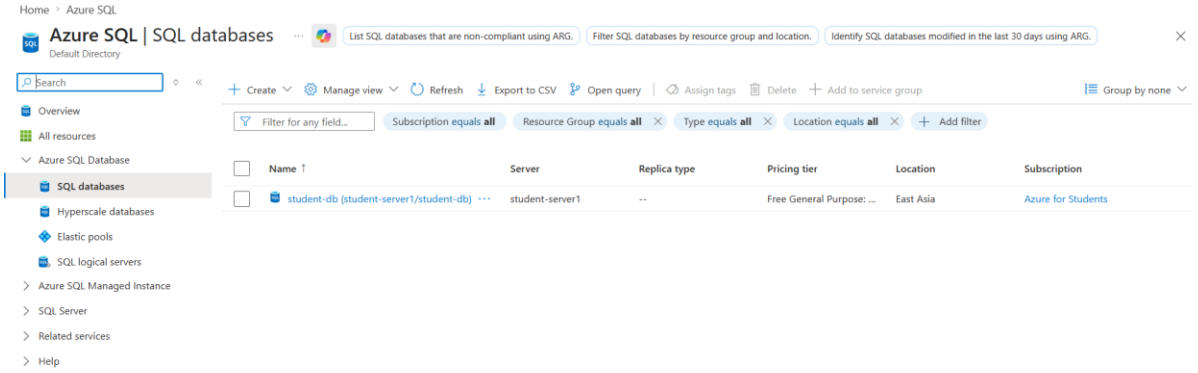
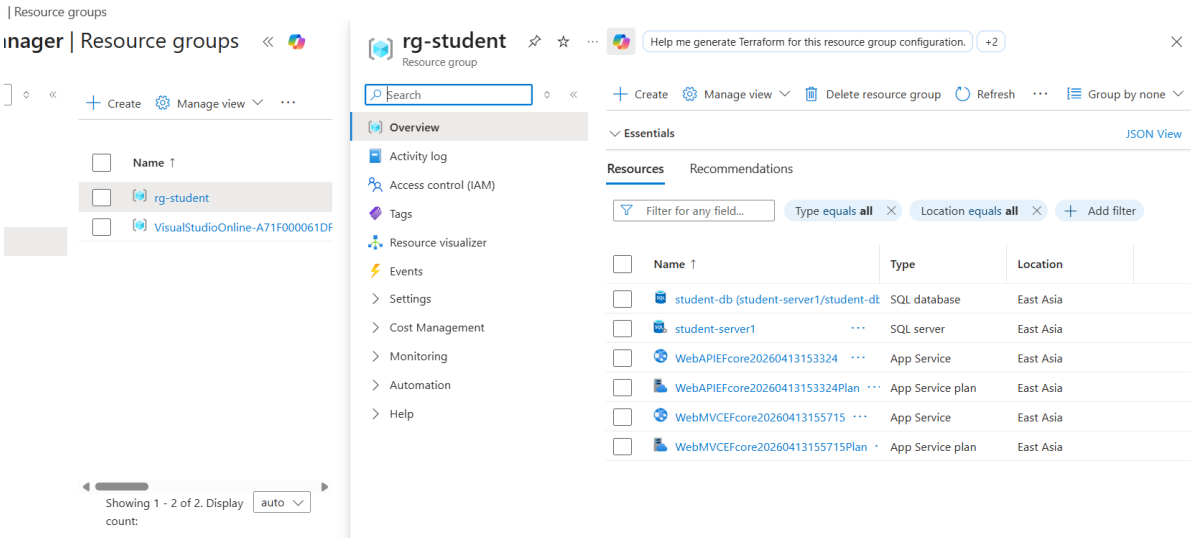
Succeeded (0 sec 831 ms) Columns: 4 Rows: 2

Pretty print

[{"id":1,"name":"Sneha Pahuja","age":21,"course":"DSA"},{"id":2,"name":"angel","age":21,"course":"WEBDEV"}]

WebMVCEFcore Home Privacy

Students

Add New

Name	Age	Course	Actions
Sneha Pahuja	21	DSA	Edit | Delete
angel	21	WEBDEV	Edit | Delete